

Case Study: Can the Weather Predict the Power Grid? Forecasting Regional Energy Demand Using Weather Patterns

DS 4002 | Pallavi Mamillapalli

Scenario

Imagine sitting in a control room at a power company on a frigid January night in Virginia. Outside, temperatures are plummeting into the teens, and millions of people are turning up their heat at the same time. Your job is to make sure the grid has enough electricity to meet current demand. Too little supply and neighborhoods go dark. Too much, and you have wasted millions of dollars of energy. The margin for error is razor-thin, and your forecast is the only thing between stability and a blackout. This case study will introduce you to time-series forecasting by providing real hourly electricity demand data from PJM Interconnection and real daily weather observations from NOAA, allowing you to build models and understand what actually drives short-term energy demand.

Context

You are a data scientist at a regional energy company that serves the Charlottesville, Virginia area. Your team has been using a standard ARIMA model, one that forecasts future electricity demand based purely on past demand values, for years. It works reasonably well, but your manager keeps asking the same question: Shouldn't the weather make our forecasts better? Intuitively, it seems obvious. People run their heat when it's cold and their AC when it's hot, but you need evidence. You have been tasked with finding out whether adding weather variables like temperature and precipitation to the model measurably reduces forecasting error, or whether the grid's own history is already capturing everything the model needs to know. The answer has real consequences for how your team allocates resources and plans for peak demand periods.

Your Task

In this case study, you will work with real data from PJM Interconnection's DataMiner2 portal and NOAA's Climate Data Online to investigate whether weather improves short-term electricity demand forecasts. You will begin by downloading and cleaning both datasets, aligning hourly load data with daily weather observations, and merging them into a single analysis-ready file. You will then build a baseline ARIMA model using only past demand values, and extend it to a model that incorporates weather as an external predictor. Finally, you'll compare the two models using RMSE and MAPE to answer the central question in a reflection: Does weather help predict weather?

All materials for this project can be found in the GitHub repo:

<https://github.com/pallavim8/Project-2-Cavalier-Kaggles>